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Modeling the effects of wave climate and sediment supply variability on large-scale shoreline change

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ABSTRACT

The application of an integrated data analysis and modeling scheme reveals that decadal-scale shoreline evolution along a U.S. Pacific Northwest littoral cell is highly dependent on both sediment supply and wave climate variability. In particular, accurate estimates of (Columbia River) sediment supply and sediment feeding from the lower shoreface are critical components of balancing the barrier beach sediment budget and are therefore essential to making sensible shoreline change hindcasts and forecasts. A simple deterministic one-line shoreline change model, applied in a quasi-probabilistic manner, enables evaluation of the influence of sediment supply and wave climate variability through simulation of historical shoreline change. Through iteration, a range of realistic scenarios are developed to constrain decadal-scale shoreline change predictions. Modeled shoreline changes are significantly sensitive to directional changes in the incident waves, and therefore sensitive to the occurrence of interannual climatic fluctuations such as major El Niño events. A predicted increase in the intensity of the east Pacific wave climate (1.0 m increase in significant wave height in 20 yr) affects forecast shoreline positions only when this increase occurs during the winter storm season. However, the effect of this increase in storm power during any given year is small relative to the impact of major El Niño events. The model has significant skill in decadal-scale hindcasts suggesting that alongshore gradients in sediment transport dominate coastal change at this scale at this site. However, both data and model results suggest that net onshore feeding from the lower shoreface is responsible for approximately 20% of the decadal-scale coastal change. Field measurements and poor model skill at annual scale indicate that cross-shore processes likely dominate coastal change at shorter time scales.
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1. Introduction

The position of the shoreline along sandy ocean coasts varies over a broad spectrum of time and space scales in response to a variety of processes (Morton, 1979; Stive et al., 2002). Shoreline position is constantly evolving toward a dynamic equilibrium reflecting a region's imposed geologic setting, sea level, sediment budget, wave climate, and hydrodynamic regime. Changes in the position of the shoreline over years to decades can be drastic and costly (damages resulting from coastal erosion in the USA alone is estimated to cost up to hundreds of millions of U.S. dollars annually, The Heinz Center Report, 2000), significantly affecting coastal communities, transportation routes, and ecosystems. Therefore, it is of paramount importance for coastal communities to know if local shorelines are stable, advancing, or retreating and how those changes are likely to manifest over larger time and space scales. Modest recent progress has been made toward developing a robust capability of forecasting annual- to decadal-scale shoreline change (Hanson et al., 2003).

1.1. Background and setting

Due to high-energy forcing and associated large scales of morphological change (Ruggiero et al., 2005; Kaminsky et al., 2010-this issue), the Columbia River littoral cell (CRLC) in Southwest Washington and Northwest Oregon, USA (Fig. 1) is an ideal natural laboratory for improving annual to decadal-scale shoreline change predictive capabilities. Since the construction of jetties in the late 1800s and early 1900s at the entrances to the Columbia River and Grays Harbor, shoreline advance has been significant, on the order of 1 km, and dominated by sediment supply from the deflation of the ebb-tidal deltas (Gelfenbaum et al., 1999; Kaminsky et al., 1999; Buijsman et al., 2001, 2003a,b). In general, shoreline advance has been greatest next

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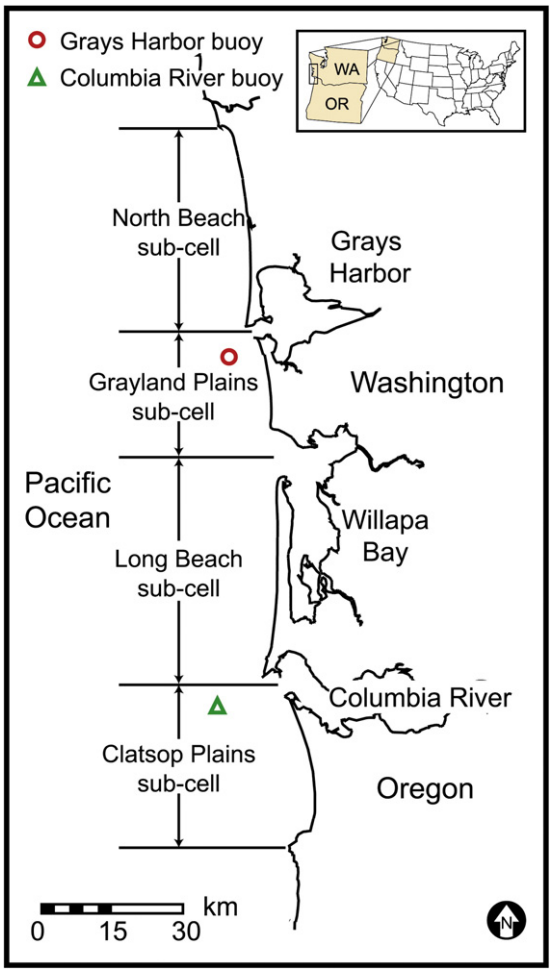


Fig. 1. The Columbia River littoral cell.

to the jetties, decreasing alongshore over tens of kilometers, resulting in curvilinear subcells (Fig. 1). Both historical bathymetric changes and the onset of coastal erosion adjacent to the jetties suggest a recent decline in sediment supply from the ebb-tidal deltas and, consequently, shoreline re-adjustment towards a new configuration.

As this long-term morphological response to jetties is occurring, a reduction in sediment supply from the Columbia River (Gelfenbaum et al., 1999) may also be reducing the continued feeding and bypassing of sediment across the Columbia River ebb-tidal delta to the adjacent coasts. Therefore, the Long Beach Peninsula subcell (Fig. 1) is likely evolving from a reduction in sediment supply from both internal (ebb-tidal delta) and external (Columbia River) source compartments of the littoral system (Kaminsky et al., 2001). Strong gradients in alongshore sediment transport rates (Buijsman et al., 2001) as well as net onshore feeding from the lower shoreface (Cowell et al., 2001; Kaminsky and Ferland, 2003) are hypothesized to dominate this evolution at decadal scales.

These decadal-scale trends are punctuated by significant seasonal variability as well as by major El Niño events that occur in the region approximately once per decade (Komar, 1986; Allan and Komar, 2002). In the Pacific Northwest, El Niños are characterized by an increased frequency of extreme waves from the south–southwest and higher than normal sea levels (Komar, 1986; Komar et al., 2000). El Niños have a well documented effect on U.S. Pacific Northwest shorelines; typically shoreline retreat occurs at the southern end of littoral compartments and progradation occurs to the north (Peterson et al., 1990; Kaminsky et al., 1998; Ruggiero et al., 2005). Changes in storm tracks, hence incident wave angles, are hypothesized to strongly affect longshore sediment transport rates and gradients resulting in these observed patterns. The influences of other factors, such as relative sea level change, may also be affecting the behavior of the shoreline but these factors appear to be of second-order relative to the dominant trends of the past century attributed to changes in sediment supply and major El Niño events (Kaminsky et al., 2001).

1.2. Approach

The objective of this study is to improve our ability to understand and ultimately predict annual- to decadal-scale shoreline change. We give priority attention here to evaluating the effect of two of the most important factors controlling shoreline change: sediment supply and incident wave climate. An integrated, multi-scale approach is taken in which decadal-scale coastal change is analyzed in context of anticipated changes at both shorter scales, i.e., seasonal to interannual shoreline variability mainly derived from beach monitoring and processes experiments (Ruggiero et al., 2005), and longer scales, i.e., coastal evolution derived from geological and historical shoreline change studies (Kaminsky et al., 1999, 2010-this issue; Peterson et al., 2010a,b-this issue). Within the context of research results over these major time scales of morphological change, a suite of data analyses and modeling tools are employed to both hindcast and forecast shoreline change (Fig. 2). The analyses of morphodynamic constituents such as waves, currents, bathymetry, topography, shorelines, and sediment characteristics yield the system sediment budget and short-term shoreline variability, both of which must be used to inform and constrain the shoreline change model. The shoreline change model itself integrates the transfer functions between fluid motion (waves and currents) and the seabed to drive both sediment transport and changes in shoreline position.

We recognize that predictions of shoreline change are inherently difficult at any scale due to the range of possible morphodynamic responses (both forced and free) to a stochastic environmental forcing (de Vriend, 1998), as well as due to a combination and interaction of processes occurring over a large range of time and space scales. In a recent review article, Hanson et al. (2003) discuss some 20 different models aimed at describing coastal change at interannual to decadal time scales, none of which presently demonstrate significant predictive skill beyond a limited range of situations. In this study we employ a simple deterministic one-line shoreline change model in

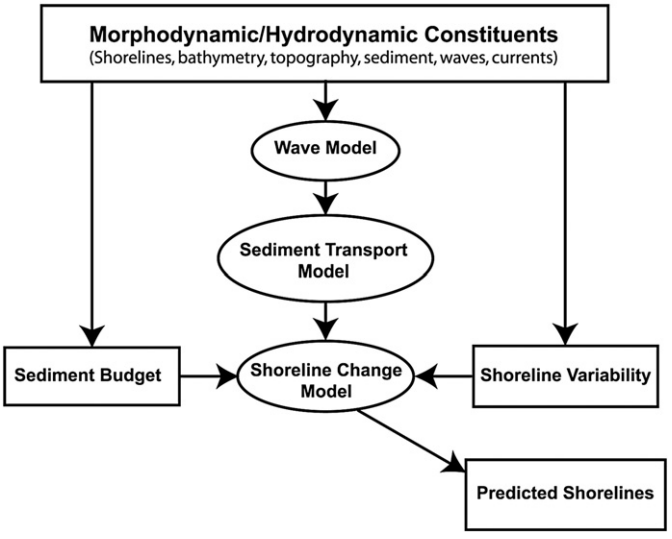


Fig. 2. Integrated analysis and modeling approach for hindcasting and forecasting shoreline change.

a region that has a strong shoreline change signal, typically strong enough to resolve net annual to decadal shoreline changes from the significant short-term fluctuations due to event- and seasonal-scale morphological variability. We apply the model in a quasi-probabilistic approach with a realistic range of possible input conditions resulting in a range of possible future conditions rather than specific shoreline change predictions. Our shoreline modeling focuses on the Long Beach sub cell (Long Beach Peninsula) within the CRLC (Fig. 1) after testing the sensitivity of the model along the North Beach subcell.

We begin by discussing the formulations (including simplifying assumptions) of the shoreline change modeling system, in particular regarding sediment transport and model sensitivity (Section 2). Next, we illustrate the abundant and high-quality data sets necessary for undertaking these modeling exercises (Section 3). Model results are given in Section 4 where we apply the model to hindcast decadal (Section 4.1) and annual (Section 4.2) scale shoreline change. Model forecasts are presented in Section 4.3 with a discussion (Section 5) and conclusions (Section 6) following. While specific model predictions are not intended to be quantitatively precise on their own, our approach allows us to examine the sensitivity of annual to decadal shoreline change predictions to environmental forcing (wave climate) and boundary conditions (sediment supply).

2. Model description

Strong gradients in longshore sediment transport rates and clear shoreline change trends (Buijsman et al., 2001; Kaminsky et al., 2010-this issue) suggest that the beaches of the CRLC are a suitable candidate for one-line shoreline change modeling (Hanson et al., 2003). One-line theory, first developed by Pelnard-Considere (1956), is based on the assumption that the cross-shore profile shape remains unchanged as the shoreline position varies. Therefore, only one contour (line) is necessary to describe the evolution of the beach planform as individual profiles are assumed to move horizontally over the entire active profile height as a result of erosion or accretion. Assuming a small incident wave angle, the equations of motion in one-line theory reduce to the well-known diffusion equation and many analytical solutions have been developed for a variety of simple situations (e.g., Bakker and Edelman, 1965; Larson et al., 1997). Recently Ashton et al. (2001) and Falques (2003) have pointed out that the classically-derived diffusion coefficient, necessary in these analytical solutions, has been significantly over-predicted. At large incident wave angles a morphodynamic instability arises owing to negative effective diffusivity. Fortunately, direct numerical solutions (e.g., Rea and Komar, 1977; Le Mehaute and Sodate, 1980; Hanson, 1989) of the one-line theory continuity equation

$$h_p \frac{dy}{dt} + \frac{dQ}{dx} + q_b = 0 \tag{1}$$

where h_p is the active depth of the profile, Q is the total longshore sediment transport rate, q_b represents local sediment sources or sinks, y is the cross-shore coordinate of the coastline position, and x is the alongshore coordinate, do not explicitly use the diffusion coefficient and are therefore not confronted with this problem.

Here we employ the quasi-2D numerical one-line shoreline change model UNIBEST-CL (WL|Delft Hydraulics, 1994), an acronym for Uniform Beach Sediment Transport, to both hindcast and forecast shoreline changes along the CRLC. UNIBEST solves Eq. (1) on a staggered grid with four options for boundary conditions; the shoreline position remains constant, the shoreline angle remains constant, the sediment transport, Q, at the boundaries remains a user-defined constant, or the sediment transport at the boundaries is a user-defined function of time. This model is typically applied to simulate shoreline responses occurring over a single year to a period of decades under imposed wave conditions, boundary conditions, initial coastline shape, and other input parameters. UNIBEST can be applied to beaches of arbitrary cross-shore profiles, with straight and parallel depth contours, which are attacked by obliquely incident random wave fields and tidal currents. In general, UNIBEST is not significantly different than other commercially available shoreline change models, e.g., GENESIS (Hanson, 1989), and therefore similar results to those presented herein could be produced by other software packages.

UNIBEST transforms incident waves from the offshore extent of individual cross-shore profiles to the shoreline, accounting for refraction, shoaling, and dissipation by wave breaking and bottom friction using a random wave propagation and decay model (Battjes and Stive, 1984). A prescribed sequence of approaching wave heights, periods, and directions (typically derived from an annualized schematization of the wave climate rather than any particular time series) is used to calculate the cross-shore distribution of wave height, wave setup, and longshore currents via the momentum equation, accounting for bottom friction, gradients in the radiation stress, and tidal currents if appropriate. Along each profile, the cross-shore distribution of longshore sediment transport is typically calculated using one of several total load sediment transport formulae such as Bijker (1971) or Van Rijn (1989). It is also possible to use the CERC formula (SPM, 1984) to calculate the cross-shore integrated longshore sediment transport rate based on the breaking wave height and angle only (no results using this formula are discussed in this paper). Using either approach, longshore sediment transport rates are interpolated along the shoreline between the profiles while gradients in transport rates, boundary conditions, and sediment sources and sinks govern shoreline change. As the shoreline progrades or retreats the shoreline orientation changes and transport rates adjust to the updated local wave approach angle. However, the cross-shore profiles remain of constant shape and simply translate horizontally as the shoreline changes.

2.1. Sediment transport formulation

Each of the total load sediment transport formulae available in UNIBEST are based on the concept that transport has a threshold of motion and that there is a maximum capacity that the flow can carry. Sediment transport is assumed to respond to the local wave and current conditions in an instantaneous, quasi-steady way. For reasons discussed below, in this paper we use the Bijker (1971) formula for bedload due to waves and currents, q_b, and a suspended load term, q_s, based on the Einstein–Rouse suspended sediment concentration profile. The total sediment transport rate per unit width along a cross-shore profile is given as

$$q_t = q_b + q_s \left(\mathrm{m^3/m/s, including\,pores}\right), \text{where} \tag{2}$$

$$q_b = b \cdot \frac{D_{50} \cdot V \cdot \sqrt{g}}{C} \cdot \exp\left[\frac{-.27 \Delta D_{50} C^2}{\mu V^2 \left\{1 + \frac{1}{2}\xi\left(\frac{u_b}{V}\right)^2\right\}}\right] \tag{3}$$

The mean flow is represented by V, the near bed RMS wave orbital velocity is given as u_b, b is a constant model coefficient (ranging between 1 and 5), D_{50} is the median grain diameter (m), and Δ is the relative density of the bed material. The Chezy roughness coefficient, C, is equal to $18 \log (12d/r_c)$ where d is the water depth and r_c is the bottom roughness equal to 0.5 to 1.0 times a user-defined ripple height. From these parameters the following bottom parameters are derived: $\mu = (C/C_{90})^{3/2}$, where $C_{90} = 18 \log (12d/D_{90})$ and $D_{90} = 90\%$ grain diameter (m), and $\xi = C\,(f_w/2g)^{1/2}$ where $f_w = \exp[-5.977 + 5.213(u_b/\omega r_c)^{-0.194}]$ and ω is the wave angular frequency (rad/s). The model is applied with bed load taking place within a layer with

thickness r_c above the bed. Above this layer the suspended load transport is given as

$$S_s = 1.83 S_b \{I_1 \ln(33d / r_c) + I_2\} \quad (4)$$

where:

$$I_1 = R \int_{r_c/d}^{1} [(1-y)/y]^{z*} dy \quad (5)$$

$$I_2 = R \int_{r_c/d}^{1} \ln y [(1-y)/y]^{z*} dy \quad (6)$$

$$R = \frac{.216 (r_c/d)^{z_*-1}}{(1-r_c/d)^{z_*}} \quad (7)$$

$$z_* = w \Big/ \left(\kappa v_*\right) \; (\kappa = 0.4) \quad (8)$$

$$v_* = g^{1/2} \frac{v}{C} \left\{ 1 + \frac{1}{2} \left(\xi \frac{u_b}{v} \right)^2 \right\}^{\frac{1}{2}} \quad (9)$$

The sediment characteristics are given as ρ_s = sediment density (kg/m^3) and w = sediment fall velocity (m/s).

2.2. Model sensitivity

Before applying UNIBEST to examine the influence of wave climate and sediment supply variability on shoreline change, a detailed model sensitivity analysis is performed to determine the impacts of minor adjustments of various model parameters on shoreline change predictions (Table 1). True values for the various model parameters are never precisely known so it is crucial in applying the model to be aware of how uncertainty in model input relates to variability in model output. Using primarily standard (default) model parameter values (i.e., minimal tuning), a 'Best Guess' of the shoreline progradation from 1950 to 1995 along the North Beach subcell (Fig. 3) is developed. Model boundary conditions are determined from the integrated sediment budget developed in Kaminsky et al. (2001) with 1.0 million (M) m^3/yr of sand entering the littoral system from the south, 0.1 Mm^3/yr of sand exiting to the north, and 0.3 Mm^3/yr of cross-shore feeding from the lower shoreface (Kaminsky and Ferland, 2003). The RMS difference between the modeled 'Best Guess' shoreline and the 1995 observed shoreline along this 25 km stretch of coast is 12% of the average observed change between 1950 and 1995 (Fig. 3).

To perform the sensitivity analysis, more than 35 model runs were conducted by varying model input and tuning parameters including wave height, period, direction, number and duration of wave classes,

Table 1
Sensitivity analysis of UNIBEST model results relative to the North Beach 'Best Guess' shoreline hindcast.

Class of model input	Parameter	Changes	RMS_o (m)
Wave characteristics	Direction	$D_p \pm 2°$; $\pm 4°$	72; 147
	Height	$H_s \pm 0.25$ m	40
	Period	$T_p \pm 1$ s	2
	Wave class duration	± 7 days	16
Sediment transport formula and coefficients	N/A	van Rijn (1989)	65
	r_c	0.05 to 0.1	18
	b	1–3	4
Sediment characteristics	D_{50}	All profiles same	18
	w	Varied with D_{50}	36
Profile shape characteristics	Active height	13 m–17 m	50
	Profile shape	All profiles same	1

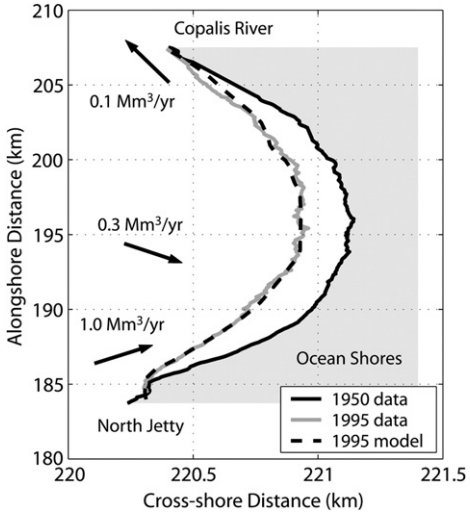


Fig. 3. North Beach model 'Best Guess' based on the sediment budget in Kaminsky et al. (2001). RMS error is 12% of the average observed change (not to scale).

sediment supply terms, depth of closure, boundary conditions, water level, sediment size, fall velocity, current velocity, bottom roughness, and calibration constants in the transport formulas. Each of the runs are compared with the 'Best Guess' shoreline position by means of an error analysis where RMS_0 is taken as the cross-shore RMS difference between the sensitivity run and the 'Best Guess'. The primary results from the sensitivity analysis are summarized in Table 1 and include:

- UNIBEST is significantly sensitive to changes in input wave direction at the offshore model boundary (approximately 16 m MLLW water depth). Rotating the wave climate used for the 'Best Guess' by $\pm 2°$ resulted in an RMS_0 value of 72 m. A rotation of $\pm 4°$ resulted in an RMS_0 value of 147 m. In contrast, variations in wave height have a relatively moderate impact while variations in wave period have a negligible impact. Increasing the duration of certain wave classes by up to 7 days, in particular those classes with relatively steep incident angles, i.e., >300° and <240°, has a moderate impact.
- Net (northward) sediment transport is almost always higher with the Van Rijn (1989) sediment transport formula than with the Bijker formula (1971), in some cases giving quite unrealistic erosion and sedimentation patterns. The application of the Van Rijn formula (1989) gives a large RMS_0 value (65 m) relative to the 'Best Guess' and therefore only the Bijker (1971) formula is used hereafter.
- In the Bijker (1971) sediment transport formulation the bottom roughness, r_c, the coefficient, b, and the criterion of breaking in the wave model, γ, are subject to tuning during model calibration. The criterion of breaking was not adjusted during the model sensitivity analysis and variations in b had a very minor influence. Only variations in the bottom roughness, r_c, significantly affects results. The bottom roughness is incorporated in the Chezy coefficient and in the bed shear stress and an increased roughness decreases sediment transport. Increasing r_c by 100% from 0.05 to 0.1 results in an RMS_0 of 18 m.
- For the 'Best Guess' simulation, grain size (D_{50}) was varied alongshore from 0.22 mm in the south to 0.12 mm in the north in agreement with sieved sand samples (Ruggiero et al., 2005). However, for this simulation the fall velocity was held to a constant (0.020 m/s). The model is moderately sensitive, $RMS_0 = 36$ m, to changes in sediment fall velocity when varied alongshore in combination with the grain size variability. The model is less sensitive to simply ignoring the alongshore variability of grain size, $RMS_0 = 18$ m.
- Using the formulas of Hallermeier (1981) and Birkemeier (1985) a range of closure depths can be calculated for the relatively intense U.

S. Pacific Northwest wave climate. Varying the active height, from 13 m to 17 m, of the individual cross-shore profiles using this range of closure depths has a moderate effect on the modeled shoreline position, $RMS_0 = 50$ m.

- Surprisingly, the detailed shape of the cross-shore profiles, including bars and troughs, has little effect on model results. Making each of the cross-shore profiles identical, rather than incorporating them into the model with measured alongshore variability, gives an RMS_0 value of 1 m.

2.3. Model assumptions and limitations

Coastal morphodynamic systems exhibit properties of non-linear dissipative dynamical systems (de Vriend, 1998). In particular they exhibit Markovian behavior in that past shoreline position strongly influences contemporary coastal change, which leads to a variety of possible future configurations. Coastal change proceeds at virtually all scales and coastal change modelers are confronted with fundamental limits of predictability related to the particular scales being modeled (de Vriend, 1998). In addition, specific limitations associated with one-line shoreline change models have been discussed at length by several researchers (e.g., Hanson, 1989; Young et al., 1995; Thieler et al., 2000). The primary assumptions made by most one-line models (including UNIBEST) include a lack of temporal evolution of the bottom profile (only longshore transport is explicitly accounted for) and the use of an active zone concept from the berm crest to the depth of closure which does not change. These assumptions lead to intrinsic model limitations including; not all of the processes that move sediments are represented; alongshore morphological heterogeneity is not accounted for; and sediment transport non-linearities, particularly associated with higher-order velocity moments, are not typically modeled. In light of these limitations our approach here is to use the shoreline change model primarily as a tool for understanding the drivers of large-scale coastal evolution, not as a deterministic method of making specific shoreline change predictions.

3. Development of model input

The model is developed to hindcast and forecast shoreline change along the Long Beach Peninsula between the rocky headland boundary at North Head and the distal end of the spit at Leadbetter Point (Fig. 4). Model input consists of beach profiles (including nearshore bathymetry), sediment grain size and fall velocity, shoreline position, representative wave climate, as well as a detailed sediment budget for determining boundary conditions. In the case of modeling shoreline change along this section of the CRLC, the subject of a major multi-year multi-agency investigation (Gelfenbaum and Kaminsky, 2010-this issue), these input data are developed with an unusual degree of quality and accuracy.

3.1. Beach profiles/sediment characteristics

Broad surf zones with multiple sandbars characterize the dissipative, infragravity energy dominated nearshore zone of the Long Beach Peninsula. The beaches are wide and gently sloping (mean foreshore slope ~1V:50H) with fine to very fine sands (alongshore averaged $D_{50} = 0.18$) having been derived from the Columbia River. The sediment size (and beachface slope) decreases northward from the Columbia from approximately 0.21 mm (1 V:37H) to approximately 0.16 mm (1 V:71H) at the tip of the peninsula, a finding that suggests net northward alongshore sediment transport (Ruggiero et al., 2005).

To account for the curvature in the shoreline and bathymetry, five cross-shore beach profiles (black horizontal lines in Fig. 4) are positioned along the shoreline with a spacing varying between 7 and 10 km. The water depth at the offshore water boundary of these profiles is approximately 16 m below mean lower low water (MLLW).

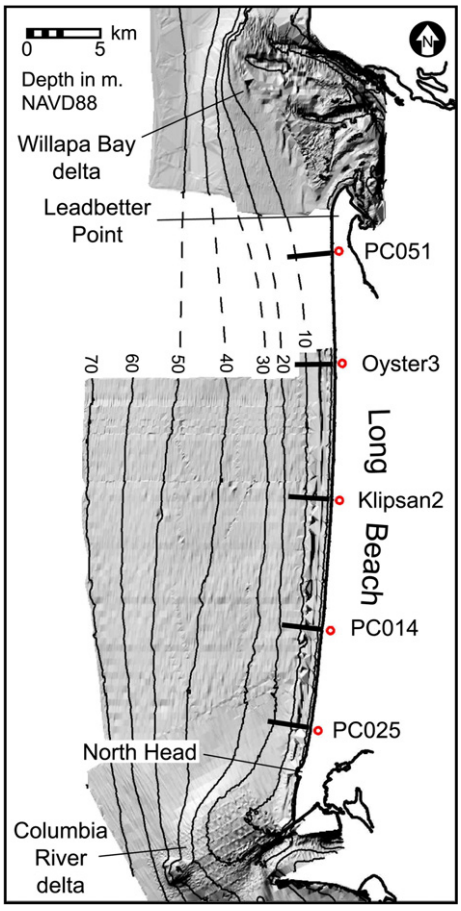


Fig. 4. The Long Beach subcell with profile locations (black lines). The dashed contour lines are estimates at locations where there are gaps in the bathymetric data. Water depths are in meters relative to mean lower low water (MLLW).

Since the sensitivity analysis indicated that model results are relatively insensitive to alongshore varying profile shape and grain size we use a single representative measured beach profile (profile PC014) and a single representative grain size ($D_{50} = 0.19$ mm) at each location.

3.2. Historical shoreline change and a quantitative sediment budget

Historical shoreline change along the Long Beach Peninsula has been determined using High Water Line shorelines derived from late 1800s to mid 1900s maps and charts and aerial photos since the 1950s (Kaminsky et al., 1999, 2010-this issue) as well as datum-based shorelines from GPS surveys and LIDAR (Ruggiero et al., 2003a; Sallenger et al., 2003). Decadal- to century-scale regional shoreline change rates reveal the initial large perturbations resulting from jetty construction (approximately the turn of the 20th century) as well as the subsequent regional-scale trends toward equilibrium (Kaminsky et al., 2010-this issue). Historical shoreline positions from the 1950s to 1995 (Fig. 5) are used for decadal numerical model hindcasts. During this period the Long Beach Peninsula shoreline experienced an average of 165 m of progradation ranging from approximately 50 m in the south to over 250 m in the north.

A large-scale morphodynamic conceptual model of sediment sources, sinks, and pathways (Kaminsky et al., 1997; Gelfenbaum et al., 1999) has been converted into a quantitative dynamic system wide sediment budget (Kaminsky et al., 2001) from detailed analyses of bathymetric, topographic, and shoreline change data for the period 1950s to 1990s (Gibbs and Gelfenbaum, 1999; Buijsman et al., 2003a). Bulk volumes and net sediment fluxes between sediment

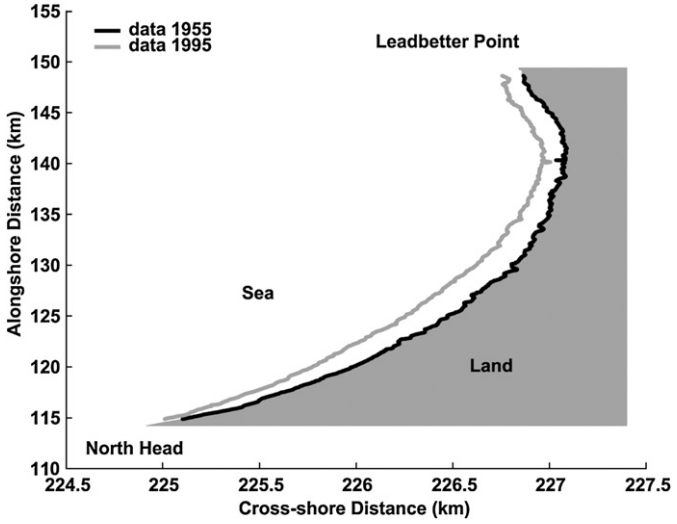


Fig. 5. Observed shoreline change along the Long Beach Peninsula between 1955 and 1995 (not to scale). The cross-shore is significantly exaggerated in order to visualize the shoreline change signal along this 35 km stretch of coast.

compartments are derived for the inner shelf, including the Columbia River inlet and ebb-tidal delta, and the adjacent coastal barriers. This quantitative sediment budget forms the basis for constraining the ranges of boundary conditions used in the shoreline change model.

Between 1955 and 1995 the Long Beach Peninsula accreted at a rate of 2.3 Mm^3/yr, totaling 92 Mm^3 over the 40 yr (Kaminsky et al., 2001; Buijsman et al., 2001). The source of much of this sand was the Columbia River ebb-tidal delta which was still responding to the construction, and subsequent maintenance, of the Columbia River jetties. Detailed bathymetric change analysis at the Columbia River entrance (Buijsman et al., 2001 between 1958 and 1998 suggests that the outer ebb-tidal delta migrated offshore, accumulating 2.0 Mm^3/yr for a total of 81 Mm^3, while the inner delta (pre-jetty channel-mouth bar) and inlet eroded at a rate of 2.8 Mm^3/yr for a total of 113 Mm^3. The upper shoreface, shallower than 24 m along the southerly 14 km of Long Beach, accumulated 0.5 Mm^3/yr, or 22 Mm^3, over this period. Present day sediment supply rates from the Columbia River to the coast are not certain. While it is possible that none of the sand currently entering the estuary from the river is reaching the littoral environment, for our shoreline change hindcasts we use the estimate of Gelfenbaum et al. (1999) that 1.4 Mm^3/yr of sand is available from the Columbia River. The lower shoreface between the 24 m and 40 m depth contours along Long Beach eroded by as much as 30 $m^3/yr\ m^{-1}$ during this period.

3.3. Wave climatology

Wave forcing is the principal driver of sediment transport in the one-line shoreline change model. Wave data for this study are derived from the Grays Harbor directional Waverider buoy (Coastal Data Information Program, Buoy 03601) located approximately 8 km southwest of the entrance to Grays Harbor, WA in approximately 43 m of water (46°51′ N, 124°15′ W, Fig. 1) and from the 3-meter discus buoy located offshore of the Columbia River ebb-tidal delta in approximately 128 m of water (National Data Buoy Center, Buoy 46029, 46°07′ N 124°30′ W, Fig. 1). The wave record at the Grays Harbor buoy comprises wave height and period data since 1981 and directional data since 1993. The Columbia River buoy has wave height and period data since 1984 and directional data since 1996. Because the directional wave record of the Grays Harbor buoy is longer and more complete than the record of the Columbia River buoy, data gaps in the Grays Harbor record are complemented with data from the Columbia River buoy in order to develop model input. The wave record derived from these buoys reveals a CRLC wave climate characterized by summer waves with a monthly mean significant wave height (H_s) between 1 and 2 m, a peak period (T_p) between 8 and 10 s, and a direction (θ) from west to northwest (Fig. 6). Winter waves have a monthly mean H_s between 2 and 4 m, a mean T_p between 10 and 14 s, and mean direction from the west to southwest. Winter storms in the region are intense, with significant wave heights exceeding 10 m and waves approaching the coast from steep southerly angles.

Many of our model simulations use a wave climate derived from approximately seven full years (1993–2001) of directional data. This 'Baseline' wave climate is referred to as wave climate 1 (WC1) in Table 2 and is constructed as follows. Waves in each buoy record are first de-shoaled and back-refracted to deep water using linear theory and Snell's law. A combined continuous seven-year time series of H_s, T_p, and θ is constructed and subsequently shoaled and refracted back onshore (again with linear theory and Snell's law) to the offshore model boundary (~16 m MLLW) of each of the five cross-shore profiles. At each profile the time series is then discretized into wave classes with associated annual probability of occurrence (Fig. 7). Fourteen directional bins between south (180°) and north (360°) are generated with a bin width varying between 5 and 30°, along with 12 wave height classes between 0 and greater than 11.5 m with a bin width between 0.5 and 1 m. Eight wave period bins are used between 0 and greater than 18 s with a bin width between 2 and 3 s. At each of the offshore model boundaries there are approximately 100 wave conditions with a non-zero probability of occurrence (Fig. 7).

Errors in the individual directional wave measurements are assumed random and, when using a long time series, have a minimal effect on the shoreline change results. However, systematic errors in the wave climate could have a significant effect. To illustrate the effect of a consistent error in the data, wave climate 2 (Table 2) is developed by simply rotating the Baseline climate clockwise by 1°.

During the 1997/1998 El Niño, U.S. Pacific Northwest beaches experienced monthly mean winter wave heights up to 1.0 m higher than usual, and wave directions having a more southwest approach than typical (Kaminsky et al., 1998; Komar et al., 2000; Revell et al., 2002; Ruggiero et al., 2005). To test the influence of these wave variations on shoreline change we generate three additional wave climates. Wave climate 3 eliminates the 1997/1998 El Niño from the seven-year record and wave climate 4 is generated specifically for the El Niño year between 1 August 1997 and 31 July 1998 (Fig. 7). Wave climate 5 is representative of the moderate wave year between 1 August 1999 and 31 July 2000 (Fig. 6).

Two final wave climates, to be used only in shoreline change forecasts, are developed based on long-term changes in wave height and period observed by Allan and Komar (2000) who show that the waves offshore of Washington (NDBC Buoy 46005) increased 0.027 m/yr in the annual average deep-water significant wave height and increased 0.059 s/yr in the annual average peak period since 1978. We first increase the significant wave height and peak period of the entire deep-water baseline wave climate by 0.27 m and 0.59 s, respectively (wave climate 6). Finally we apply the increase in wave height and period only to winter storm waves (wave climate 7).

4. Model results

In each model simulation described below we use insight gained from the sensitivity analysis (Section 2) and the quantitative sediment budget developed for the Long Beach Peninsula (Section 3.2).

4.1. Hindcasting decadal-scale shoreline change

Shoreline change along the Long Beach Peninsula is first hindcast for the period between 1955 and 1995. Model boundary conditions

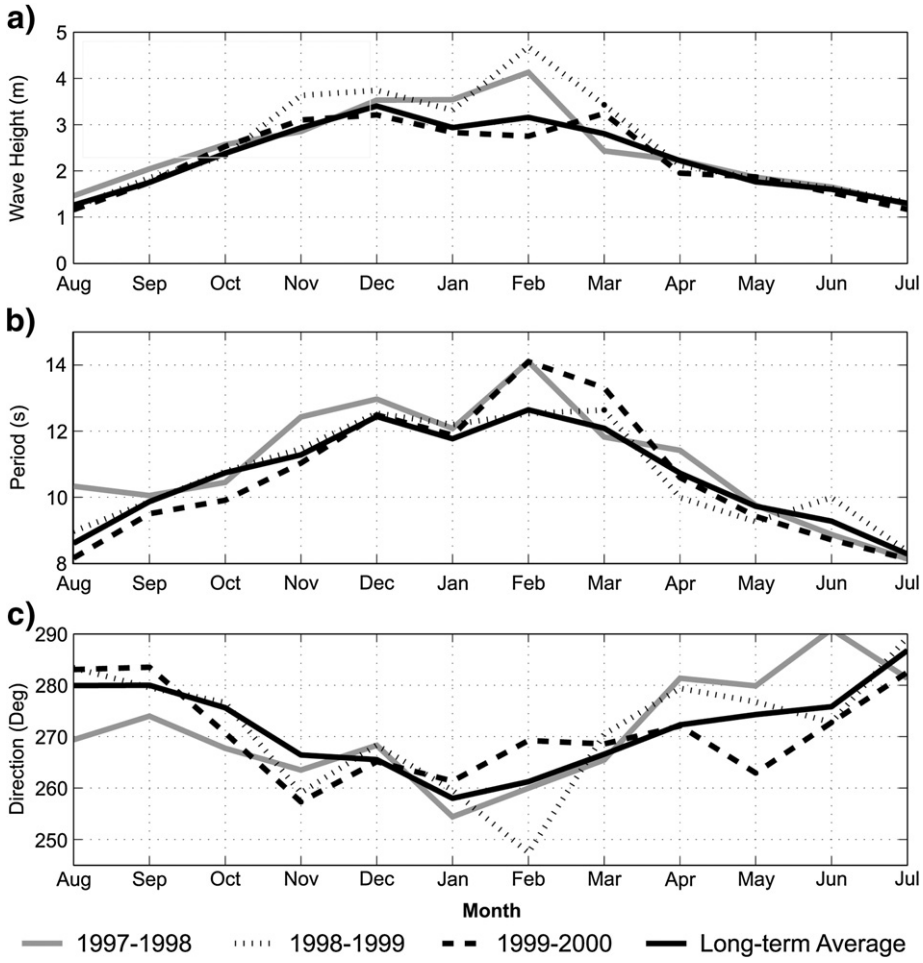


Fig. 6. Monthly mean a) significant wave height, b) peak period, and c) direction of the combined Grays Harbor and Columbia River buoy waves in deep water.

are determined through a combination of model application and results from the quantitative sediment budget. Using the Bijker (1971) sediment transport formula, the Baseline wave climate, and the shoreline angle along the northern tip of Long Beach Peninsula, the longshore sediment transport rate near Leadbetter Point is estimated to be approximately 0.4 Mm^3/yr, directed to the north. This annual sediment flux is taken as the northern boundary condition in the model (Fig. 8a). Through an iterative book-keeping process in which we keep the sediment budget compartment values of highest accuracy fixed (i.e., barrier accumulation rate), the sediment budget at the mouth of the Columbia River is balanced by allowing the remaining compartments to vary within a range constrained by data and modeling (Kaminsky et al., 2001). An annual northward littoral sediment supply of 2.3 Mm^3, feeding southern Long Beach Peninsula, is derived by this methodology and used as the southern boundary condition in the model (Fig. 8a). Due to the relatively steep angle between the 1955 shoreline and the incident waves, the longshore sediment transport rate just north of the North Head rocky promontory is significant, and calculated by the model to be approximately 1.4 Mm^3/yr directed to the north. The difference between the sediment supply boundary condition and the calculated longshore transport rate is necessary to simulate the shoreline progradation that occurred during this period. It is hypothesized that the aforementioned erosion of the lower shoreface (Section 3.2) indicates net onshore feeding, and in fact the application of a cross-shore feeding rate (from the lower shoreface) of 0.4 Mm^3/yr (~11.4 $m^3/yr\ m^{-1}$) along the subcell (Fig. 8a) is necessary to balance the sediment budget (Buijsman et al., 2003a). This volume is applied uniformly in the model as a sediment source term (boundary condition) every kilometer in the alongshore.

Table 2
Description of the seven wave climates used in a variety of shoreline change hindcast and forecast scenarios.

Wave climate	Time span	Description
1	1993–2001	Baseline climate: full time series
2	1993–2001	1° rotation of Baseline
3	1993–1997, 1998–2001	Excludes El Niño from Baseline
4	1997–1998	El Niño only
5	1999–2000	Moderate wave year
6	1993–2001	Uniform increase in H_s and T_p
7	1993–2001	Increase in H_s and T_p applied to winter season

With these boundary conditions and the Baseline wave climate, the shoreline change model is able to reproduce the measured shoreline change signal from 1955 to 1995 (Fig. 8b). The mean modeled shoreline advance is 168 m over the 35 km and the RMS error of this Baseline model hindcast (relative to the data) is 11 m. Sensitivity to the southern boundary condition, and therefore our ability to quantify the system sediment budget, is tested by increasing and decreasing the southern boundary condition by 0.5 and 1.0 Mm^3/yr. Fig. 9 illustrates the dramatic effect, over/underestimating the shoreline change signal by as much as 300 m, these changes in the sediment supply boundary condition have on the hindcast results.

The effect of variations in the wave climate relative to the Baseline model hindcast is tested by applying two of the wave climates listed in Table 2, WC2 consisting of a one degree clockwise rotation of the

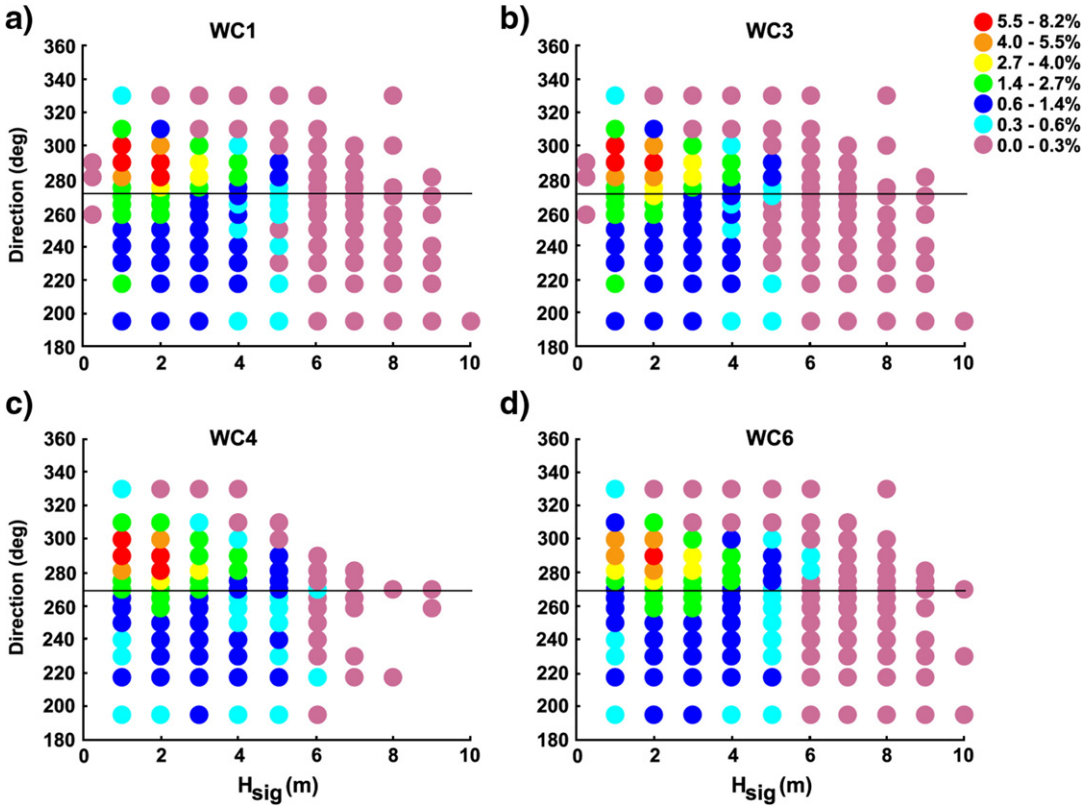


Fig. 7. Probabilities of occurrence within a particular wave height and wave angle bin of a) the Baseline (WC1) wave climate, b) WC3, c) WC4, and d) WC6 in deep water.

Baseline wave climate and WC3 excluding the 1997–1998 El Niño year. These climates are applied to the period 1955 to 1995, using the same boundary conditions as the Baseline run and the differences in the modeled 1995 shoreline position are presented in Fig. 10. As a result of the one degree rotation of the wave climate, relatively more waves approach the coast from the northern sector, generating less northward sediment transport. The sediment supply at the southern boundary is therefore considerably larger than the calculated longshore sediment transport rate, causing approximately 100 m more progradation than the Baseline results (Fig. 10b). The northward directed sediment flux near the northern boundary is smaller than that calculated for the Baseline case, and therefore with the same northern boundary condition of 0.4 Mm^3/yr to the north, the WC2 run results in approximately 100 m less progradation over the 40 yr simulation. During the Baseline hindcast model the 1997–1998 El Niño (in WC1) effectively occurs approximately six times during the simulation from 1955 to 1995. Therefore, WC3 also has significant consequences for shoreline change due to lower northward longshore

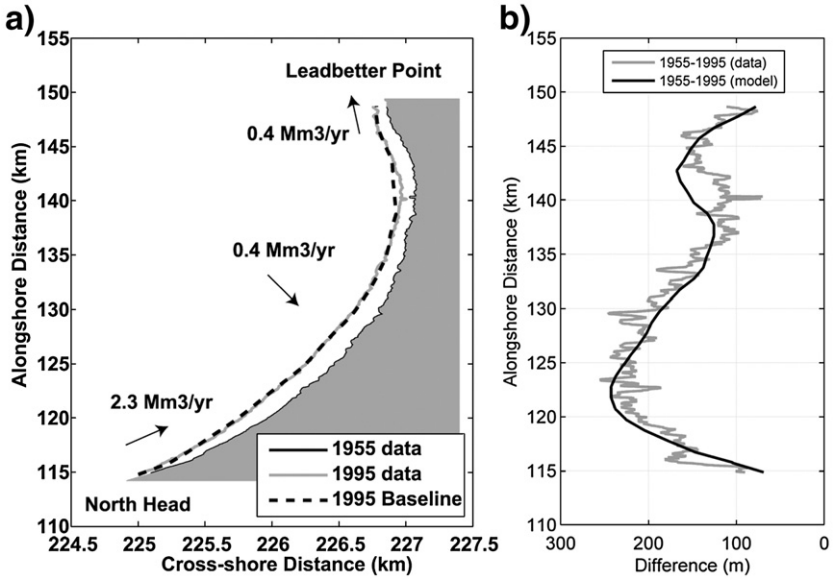


Fig. 8. Baseline model run for the period 1955 to 1995 with a) a plan view of the shoreline change and boundary conditions and b) the actual and simulated shoreline change between 1955 and 1995. The RMS error of the model results is approximately 11 m.

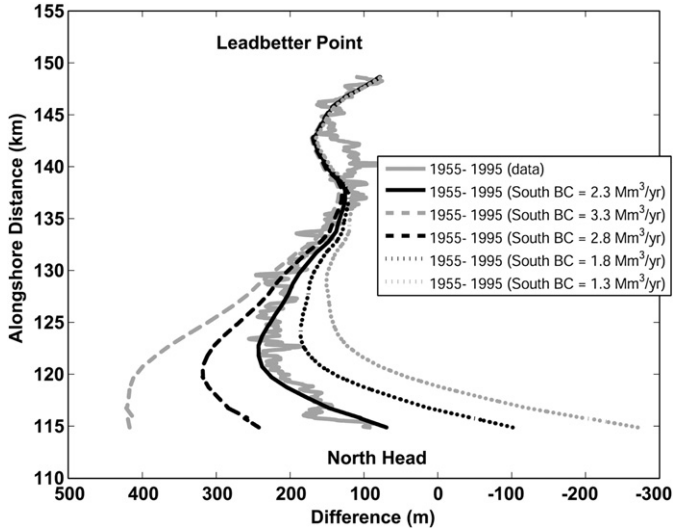


Fig. 9. Effect of varying the southern boundary condition (sediment supply) in hindcasting the Long Beach Peninsula shoreline change between 1955 and 1995.

sediment transport rates. The differences between hindcast shoreline positions using WC2 and WC3 and the Baseline hindcast are similar and shown in Fig. 10b. These results illustrate the importance of major El Niño events to decadal-scale shoreline change patterns.

4.2. Hindcasting annual shoreline change

Topographic beach measurements reveal that the 1997/1998 El Niño year resulted in as much as 30 m of shoreline retreat along the southern 25 km of Long Beach and shoreline advance of up to 50 m along the northern 10 km. This shoreline change pattern (Fig. 11a) suggests an increased northward sediment flux and mimics the counter-clockwise shoreline rotation patterns observed by Komar (1986) and Peterson et al. (1990) following the 1982/1983 event along U.S. Pacific Northwest coastlines.

To simulate the effects of the 1997–1998 El Niño, the shoreline change model is applied for a single year using WC4. To correspond with the dates of shoreline position data collection, August and September 1997 and 1998, the modeled El Niño year is defined as being between 1 August 1997 and 31 July 1998. The El Niño model run uses the same boundary conditions as the Baseline model run, except that input water levels were raised approximately 15 cm to represent the super-elevation of water levels that occurs during El Niño events. Shoreline position data, extracted from topographic beach profiles, are used to initialize the model and to compare with model results. The computed longshore sediment transport rates along Long Beach are higher than the Baseline run because of the change in the annual wave climate. This causes shoreline retreat in the southern portion of the cell and progradation to the north, similar to the observed pattern. However, the overall agreement between the model and data is poor as the large gradients in sediment transport required to cause the observed shoreline change could not be accurately reproduced by the one-line model (Fig. 11a).

The model is also used to examine shoreline change during 1999 to 2000, a year with a more typical wave climate (Fig. 6, WC5) but an atypical shoreline change signal. During the year from 1 August 1999 to 31 July 2000 the topographic beach profile data reveal shoreline advance averaging 23 m, with a maximum of 60 m, along the coast of Long Beach (Fig. 11b). The model is again initialized with the Baseline boundary conditions and once again the model–data agreement is quite poor. The model significantly under predicts the shoreline progradation, averaging only 5 m of advance.

4.3. Forecasting decadal-scale shoreline change

Kaminsky et al. (2001) demonstrate how trends in bathymetric change data can be used to develop future sediment budget scenarios. Extrapolating the accumulation and erosion trends of the Columbia River outer and inner ebb-tidal delta/inlet respectively, results in a reduction of the sediment flux feeding the Long Beach subcell after 1995 by approximately a factor of two from the historical (1955–1995) rate. Assuming a future (after 1995) sediment supply at the southern boundary condition of 1.15 Mm^3/yr, decadal-scale shoreline change predictions, for the period from 1995 to 2020, are made using several wave climates and the 1995 Baseline shoreline as a starting point. The cross-shore feeding term (0.4 Mm^3/yr) and the sediment flux out of the system to the north (0.4 Mm^3/yr) are held the same as for the model hindcasts.

The 2020 shoreline change prediction for WC1 (Baseline) suggests shoreline retreat of approximately 200 m within 5 km north of North Head (Fig. 12). This result implies that the northward directed

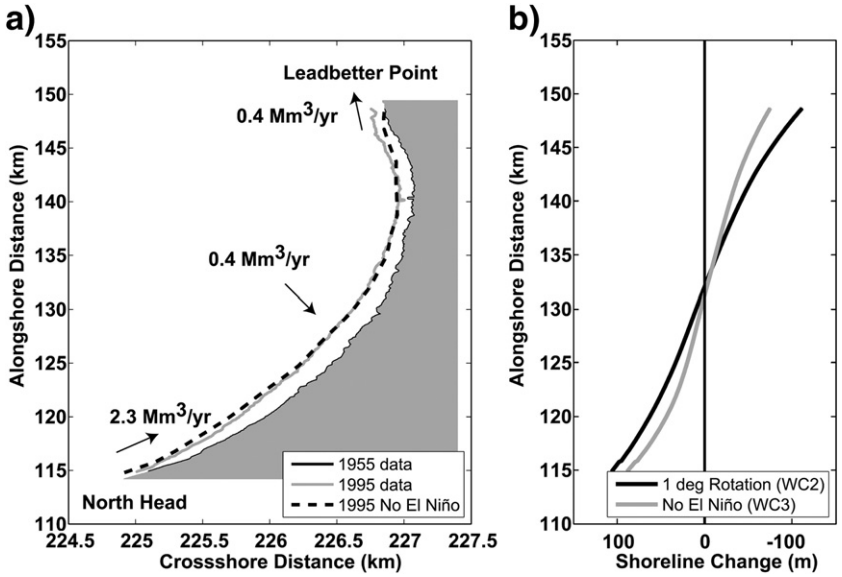


Fig. 10. a) Shoreline change hindcast using WC3 (no El Niño) as compared to data, and b) the difference between the 1995 Baseline (WC1) and both the 1995 WC2 and WC3 model hindcasts.

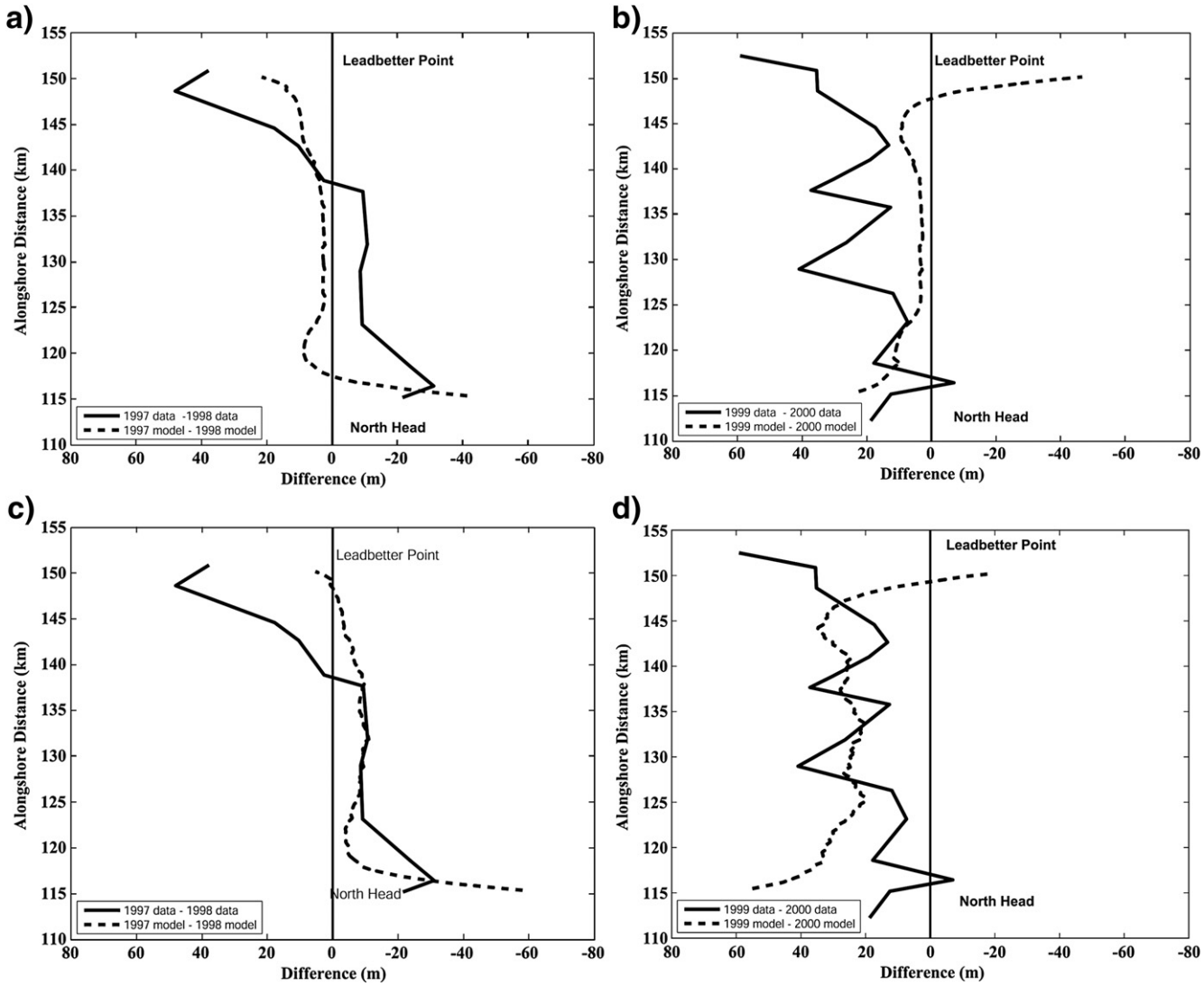


Fig. 11. Model results for simulation of a) the 1997–1998 El Niño and b) the more typical wave year 1999–2000 with a 0.4 Mm^3/yr cross-shore feeding term. A 4.7 Mm^3/yr cross-shore loss is applied to the c) 1997–1998 El Niño model run and a 9.5 Mm^3/yr cross-shore feeding term is applied to the d) 1999–2000 model run to illustrate the importance of cross-shore processes at annual scale. Shoreline advance is positive; shoreline retreat is negative.

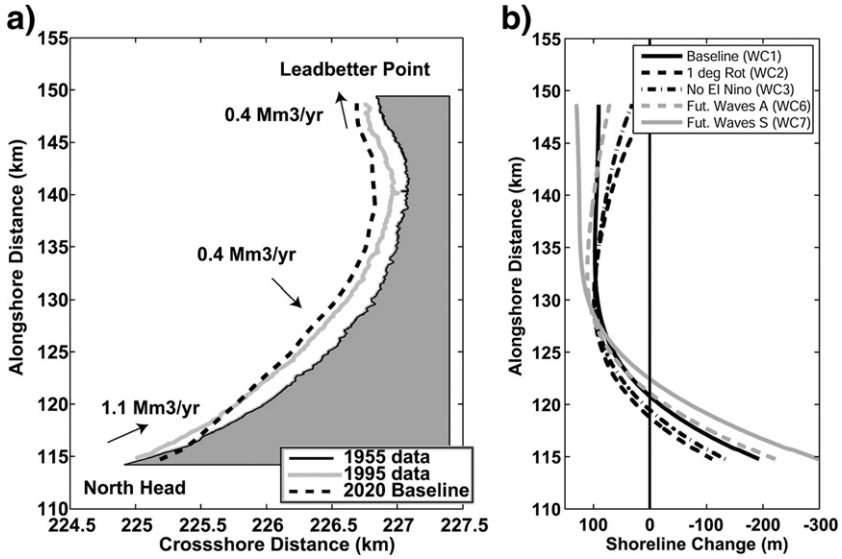


Fig. 12. a) Baseline shoreline change forecast for period 1995 to 2020 using WC1, and b) predicted shoreline change between 1995 Baseline model results and 2020 shoreline forecasts for all scenarios. Accretion is positive, erosion is negative.

sediment transport rate at the southern end of the model is higher than the estimated southern sediment supply boundary condition. North of this point Long Beach continues to accrete, with a maximum shoreline advance of up to 100 m occurring between km 130 and 150. The simulations using WC2 and WC3 are similar, showing less shoreline retreat compared to WC1 north of North Head, and less shoreline advance south of Leadbetter Point (Fig. 12b). The WC6, generic future increase in wave height and period, predictions are also fairly similar (Future Waves A in Fig. 12b). Apparently, the effect of the increase of the wave height and period does not significantly increase the net longshore sediment transport gradients as compared to the Baseline climate. However, when applied to the winter storm season alone (Future Waves S in Fig. 12b), the increased wave conditions magnify the Baseline forecast trends. These results in combination suggest that a regional trend of shoreline rotation is likely to occur over the next two decades.

Shoreline position data derived from the beach monitoring program provides information that can be used to refine the southern boundary condition to be used with model forecasts. Between 1995 (proxy-based shoreline derived from aerial photos) and 2004 (datum-based shoreline derived from topographic beach profiles) the Long Beach Peninsula averaged approximately 30 m of progradation (Fig. 13). However, in contrast with the solely progradational shoreline change signal between 1955 and 1995, significant shoreline retreat (as much as 35 m) occurred over the southern 5 km of the subcell. The shoreline prediction, as of 2004, using the sediment supply value derived from the future sediment budget scenario (1.15 Mm^3/yr) does a reasonable job of predicting the progradation over the northern part of Long Beach but over predicts the erosion at the southern end of the subcell (Fig. 13). The RMS error between model and data using the boundary condition solely derived from bathymetric change data is 27 m. Through iteration it is apparent that by increasing the sediment supply at the southern boundary to 1.5 Mm^3/yr significantly improves the data–model comparison (RMS = 21 m). Particularly noteworthy is the match of the modeled transition region between progradation and retreat and that of the data (Fig. 13). The 0.35 Mm^3/yr difference between our original estimate of sediment supply and our calibrated estimate of sediment supply is within the range of errors inherent in bathymetric change analyses and the assumptions necessary to derive a boundary condition.

Also shown in Fig. 13 is a nine-year shoreline change forecast using the sediment supply term used for the decadal-scale hindcasts,

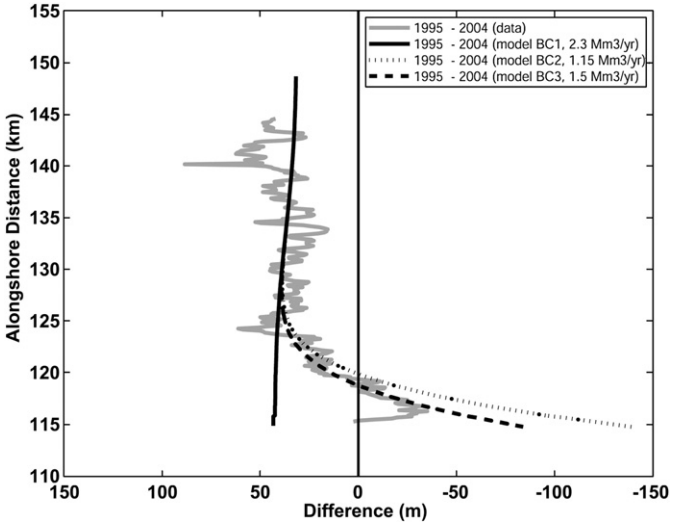


Fig. 13. Nine-year model forecast used to tune the southern sediment supply boundary condition. A southern boundary condition of 1.5 Mm3/yr does a good job of reproducing the observed shoreline change.

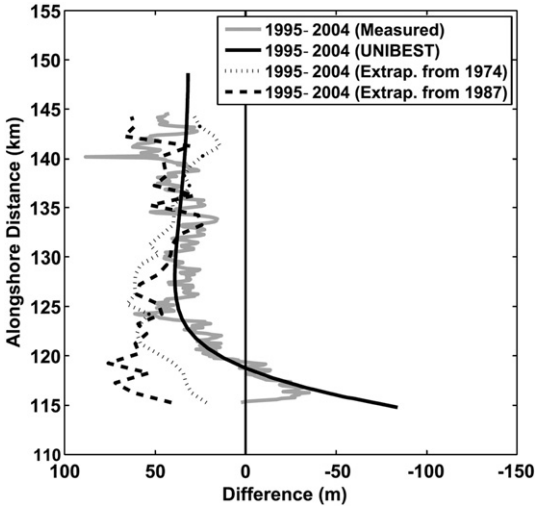


Fig. 14. Nine-year model forecast using 1.5 Mm3/yr as the southern sediment supply boundary condition (red line) as compared to shoreline change forecasts developed by linear extrapolation of the shoreline change trend between 1974 and 1995 (dotted line) and between 1987 and 1995 (dashed line).

2.3 Mm^3/yr. The prediction of shoreline progradation in the southern part of the subcell, rather than the observed retreat, using this boundary condition strongly supports the results of the bathymetric change analysis which indicates a reduction in sediment supply after 1995. The importance of understanding the temporal evolution in sediment supply to the system is further illustrated by comparing model forecasts with simple shoreline change predictions based on extrapolations of historical shoreline change trends. Fig. 14 shows two nine-year shoreline forecasts based on linear extrapolations from 1995, one using the shoreline change trend between 1974 and 1995 and the second using the shoreline change trend between 1987 and 1995. Neither extrapolation-based shoreline change forecast predicts the observed erosion in the southern few kilometers of the subcell.

5. Discussion

The shoreline change predictions, over the period 1995 to 2020, for each of the 5 wave climate scenarios examined in Fig. 12 suggests shoreline retreat of between 125 and 300 m within 4–6 km north of North Head. North of this point the Long Beach Peninsula is predicted to continue to prograde, with a maximum shoreline advance of approximately 100 m occurring around km 130–135. The consistency of the predictions lends confidence in the results. The application of our integrated analysis and modeling scheme to a variety of wave climate and sediment budget scenarios allows us to work towards applying the model in a quasi-probabilistic mode. Probabilistic modeling techniques (as opposed to strictly deterministic approaches) are appropriate for coastal change because coastal systems are non-linear with random forcing. A probabilistic approach reveals how both variability and uncertainty in input conditions is transferred to a range of predicted shoreline positions. A probabilistic approach to shoreline change modeling also allows iteration through a variety of proposed sediment management plans currently being considered by decision makers who manage dredging and disposal of sand at the mouth of the Columbia River.

A decreasing supply of sediment has been shown to have a significant impact on shoreline change predictions along the Long Beach Peninsula (Fig. 13). As an extreme case, if the sediment supply from the Mouth of the Columbia River is assumed to go to zero due to river regulation and the removal of available sand by dredging, hundreds of meters of shoreline erosion is predicted over the next several decades (not shown). It is, however, difficult to know if and

when sediment supply to the Long Beach Peninsula might equal zero without a detailed analysis of larger-scale regional trends toward equilibrium, especially related to shoreface — shelf evolution and estuary evolution. Both the lower shoreface and the estuaries are potentially significant source or sink terms that will continue to affect the natural sediment supply to the barrier beaches. In addition, appropriate dredging and disposal practices may have a significant (positive) influence on the sediment supply.

The possibility of net sand supply from the lower shoreface to beaches has been hypothesized to occur in a wide variety of coastal environments (e.g., Stive et al., 1999; Cowell et al., 2001; Kaminsky and Ferland, 2003; Aagaard et al., 2004), and based upon the observed erosion of the lower shoreface off of the Long Beach Peninsula we have made this same hypothesis here. However, direct evidence for net shoreface feeding has been elusive due to the difficulty in measuring cross-shore transport in the field, particularly over time scales long enough to capture anything close to the full climatology of flows driving sediment transport. Cowell et al. (2001) summarize several convergent lines of evidence including long-term bathymetric change analysis (such as in our case), measurements of sediment transport on the shoreface, and modeling (both behavior and process-based) of shoreface sediment transport that all indicate that sand supply from the shoreface is widespread. From a process-based perspective it has been hypothesized that during energetic conditions, such as the waning stages of storms when the downwelling associated with wind forcing abates, wave-asymmetry induced sediment transport is primarily onshore directed (Stive et al., 1999). Aagaard et al. (2004) suggest a possible mechanism for shoreface sediment transport through the surf zone to the inter-tidal where sand then becomes available for dune/barrier building. They hypothesize that a combination of relatively large onshore transports due to asymmetric incident waves and relatively small undertow velocities (occurring on low sloping beaches during surges) at times of high energy results in persistent onshore transport. While the evidence for shoreface supply to beaches in the CRLC is inconclusive, the detailed sediment budget analysis performed in Kaminsky et al. (2001) and Buijsman et al. (2003a,b) demonstrates that only part of the large progradation rate of the Long Beach Peninsula can be accounted for through direct sand supply from the Columbia River and through the degeneration of the ebb-tide delta. We note here that the reasonably good hindcast model skills (Figs. 8 and 13) discussed in this paper could not be reproduced without the inclusion of the shoreface feeding boundary condition.

The approach of using a realistic range of sediment supply scenarios with various wave climatologies for developing a 25-year forecast along the Long Beach Peninsula is illustrated by fifteen individual model runs in Fig. 15. The five wave climates (WC1, WC2, WC3, WC6, and WC7) used to generate Fig. 13 are here used in combination with three sediment supply scenarios; a beneficial use of dredged material scenario (1.75 Mm^3/yr), a do nothing scenario (1.5 Mm^3/yr), and a sediment supply reduction scenario (1.0 Mm^3/yr). The mean and standard deviation of the model forecasts are presented giving a quasi-probabilistic sense of the expected range of future shoreline conditions along the Long Beach Peninsula. The mean model forecast suggests erosion of approximately 150 m in the south and 65 m of progradation at the northern boundary by 2020. However, the results near both boundaries have the highest standard deviation and therefore the confidence in the magnitudes of coastal change at the boundaries is the smallest. The approximately 100 m of progradation near kilometer 130 is consistently predicted by virtually every model run and confidence in this result is high. It should be noted that we are not explicitly modeling the effects of the North Head rocky promontory. North Head will likely exert geologic control over the extreme southern end of the model domain locally limiting the potential for significant erosion.

Contemporary morphological changes along the length of the Long Beach Peninsula are documented through a beach morphology

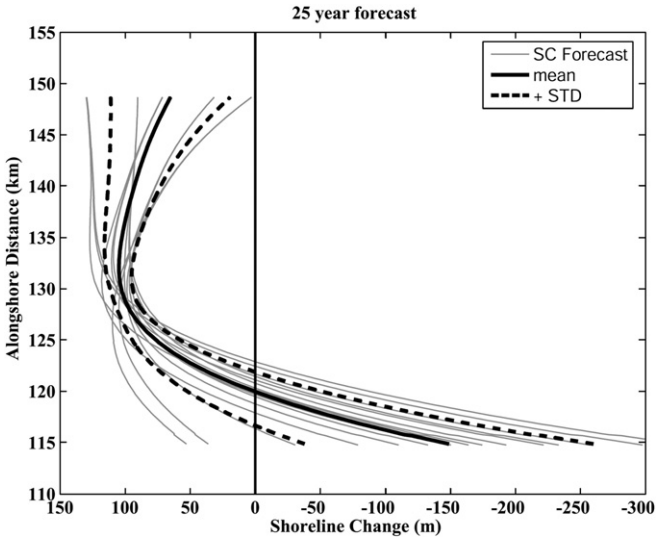


Fig. 15. Decadal-scale shoreline change forecasts resulting from 15 different combinations of sediment supply and wave climate (gray lines). The solid black line is the mean shoreline change prediction and the dashed black lines are ±1 STDEV from the mean.

monitoring program (Ruggiero et al., 2005). Three-dimensional topographic beach surface maps, cross-shore beach profiles, nearshore bathymetric profiles, and surface sediment samples quantify the short- to medium-term (event–seasonal–interannual) variability of the coastal planform and shoreline. The sub-aerial beachface typically lowers approximately 0.5 m and retreats horizontally, at the 3.0-m contour (NAVD88), approximately 20 m each winter. While these beaches tend to fully recover during the relatively low-energy summer, the seasonal variability is typically significantly greater than annual to interannual scale shoreline change. Through a combination of data analysis and cross-shore profile modeling, Ruggiero et al. (2003b) suggest that at short time scales (seasonal) cross-shore sediment transport processes are more important than alongshore sediment transport processes in reproducing observed seasonal morphological change. The annual shoreline change simulations presented in Fig. 11 indicate that from 1997 to 1998 and from 1999 to 2000 modeled gradients in alongshore sediment transport are also not sufficient to reproduce annual shoreline change. Cross-shore (surf zone) processes, which are not included in the model, probably play an important role in the observed annual shoreline change.

The 1997–1998 El Niño wave climate (WC4) forced higher than typical longshore sediment transport rates resulting in shoreline rotation, but gradients in longshore processes alone did a poor job of reproducing the observed shoreline change. In order to simulate a loss of sand from the active profile to the shelf, a sink term (estimated from measured nearshore bathymetric change) was added at 1-km intervals in the alongshore direction. The simulated loss is equal to about 130 $m^3/yr\ m^{-1}$ (4.6 Mm^3/yr for a stretch of 35 km). This loss is also equivalent to approximately a 10-cm profile lowering over a cross-shore length of 1500 m, a value of the same order of magnitude as observed losses (not shown). Adding this cross-shore sink term improves the model fit over much of the model domain, but the fit in the north is still poor (Fig. 11c).

It is hypothesized that the cross-shore losses associated with the El Niño year were approximately doubled by additional cross-shore losses during the La Niña year of 1998–1999, a second consecutive stronger than typical wave year (Fig. 6). This material was not transported back onshore until the first subsequent calm year, 1999–2000. Applying a sediment source term of 260 $m^3/yr\ m^{-1}$ (approximately 9.2 Mm^3/yr over the 35 km) more accurately reproduces the observed shoreline progradation during the moderate wave year (Fig. 11d). These imposed cross-shore fluxes account for

approximately 70% and 80% of the total annual coastal change for the El Niño year and the moderate year respectively. In contrast, on decadal scales over 80% of the accumulation of Long Beach Peninsula can be attributed to gradients in longshore sediment transport while only approximately 20% is thought to come from onshore directed cross-shore fluxes.

6. Conclusions

The combination of high-quality input data, a quantitative sediment budget for determining boundary conditions, and large coastal change signals makes the Columbia River littoral cell an ideal location for developing and testing coastal change models. In this paper, a deterministic one-line shoreline change model is applied in a quasi-probabilistic manner to test the effects of wave climate and sediment supply variability on decadal-scale hindcasts and forecasts. Our modeling exercises indicate that shoreline change is very sensitive to wave direction changes. A rotation of the wave climate results in the rotation of the equilibrium shoreline configuration in the same direction. The shoreline change model results are therefore sensitive to major El Niño events in which incident wave angles are anomalous. The effect of an increasingly intense future wave climate depends on whether the increase in the severity of wave conditions is distributed throughout the entire year or solely during the winter storm season.

Model results derived with a one-line shoreline change model are not necessarily unique as it is possible to arrive at similar results with slightly different wave climatologies and boundary conditions. However, application of the model allows for iteration with detailed sediment budget scenarios, and ultimately provides insight to the processes responsible for shoreline change. Through this iterative data analysis and modeling approach, hypotheses explaining coastal evolution are tested. For example, the hypothesis that the rapid (decadal-scale) growth of the Columbia River littoral cell barriers is a result of a combination of strong gradients in longshore sediment transport and net onshore feeding from the lower shoreface is supported by our modeling exercises. While at decadal-scale longshore processes are primarily responsible for coastal change, it is hypothesized that cross-shore processes dominate at annual scale, particularly during major ENSO cycles. Large El Niño events with high waves and relatively steep southerly wave angles are clearly responsible for large-scale-shoreline orientation adjustment, while significant cross-shore transport may be due to increased water levels. The effect of increased water levels (sustained over a winter season) on cross-shore transport in the CRLC needs further investigation.

Accurate predictions of shoreline change ultimately depend on accurate estimates of the sediment budget and representative wave climate. The future wave climate will remain difficult, if not impossible, to predict but the statistical representation of the CRLC wave climate should improve over time as the wave record is extended. Future sediment budgets must rely on detailed analyses of bathymetric and topographic change combined with sediment transport modeling along with an assessment and knowledge of longer-term equilibrium tendencies of the regional system. As comprehensive modeling capabilities of shelf to shore sediment transport remains at a rudimentary level, as does modeling that links sediment transport between the shelf and estuary, a key challenge is to improve such models.

Acknowledgements

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